Week 1 – Workshop
Gradients, Curvature, and Potential-Field Intuition

Geophysics 3A: Potential Fields & Geothermics

July 6, 2026

Duration: 3 hours **Setting:** Workshop room

Preparation

Pre-reading: Course Notes Ch. 1, 3–4
Lecture videos:
Notes:

Purpose

The purpose of this workshop is to build an intuitive and practical understanding of gradients and curvature as tools for interpreting geophysical fields. Using familiar examples from topography, you will connect physical intuition (slope, steepness, and direction) to their mathematical descriptions (partial derivatives, gradients, and the Laplacian), and begin to see how these ideas extend to abstract potential fields used in geophysics.

This workshop also introduces back-of-the-envelope (BOTE) reasoning as a core skill for estimating physical quantities, assessing plausibility, and developing quantitative intuition before applying more formal calculations or computational methods.

Learning Outcomes

By the end of this workshop, you should be able to:

- Describe the gradient of a scalar field in both intuitive (slope and direction) and mathematical (partial derivatives and vector representation) terms.
- Interpret maps and surfaces by identifying regions of high gradient magnitude and explaining their physical significance.
- Conceptually explain curvature (Laplacian) as changes in slope and identify where it is large or near zero in maps or surfaces.
- Construct and interpret gradient vectors graphically from contour maps.
- Apply simple back-of-the-envelope calculations to estimate physical quantities and justify assumptions using order-of-magnitude reasoning.
- Connect the concepts of gradients and curvature to the behaviour of potential fields and their role in geophysical interpretation.

Session Flow (170 min)

0–15 min: Course introduction

What the course covers, weekly rhythm, assessment structure, expectations, office hours, and where to find materials. Download mateials while instructor goes over outline of course.

15–60 min: Activity A: Tectonic Plate vs. Freight Train

Work in groups of 3–4 on the back-of-the-envelope style problem.

To estimate these parameters, you will need to know/learn a few things about the structure of Earth, the physical properties of common rock types within each layer, and make assumptions (understand ranges) for required values. You will need to keep track of units and become comfortable with uncertainty and order-of-magnitude estimation.

60–80 min: Mini-lecture and Q & A

Address questions from lecture; otherwise give a concise teaser: scalar vs vector fields, ∇h as direction of steepest increase, and ∇^2 as curvature, vector field = gradient of scalar potential ($F = \nabla\phi$).

80–85 min: Break

85–100 min: Activity B: Sheet Model of Potential Fields

Hands-on demonstration of a scalar field using a stretched sheet. Boundary heights are fixed (Dirichlet boundary conditions), while pushing up/down on the sheet introduces localized sources and sinks.

Focus on how the gradient (slope) and curvature (Laplacian) respond to boundary conditions and internal forcing, and how increasing observation distance smooths the field.

100–130 min: Activity C: Gradient Activity

Work in groups of 3–4 on the gradient activity.

130–145 min: Activity Debrief

Discuss the answers to the Gradient Activity.

145–155 min: Preview: Physical properties

Next week, we'll examine what I like to call the *geophysical fudge factors*: the physical properties of Earth materials. These properties control how fields are generated, modified, and transmitted. Mathematically, they often appear as scale factors that determine magnitudes and ensure the units make sense—but physically, *properties contain the geology*.

As geophysicists, we measure fields but interpret physical properties, so it is essential to understand how these properties vary with composition and state variables such as pressure and temperature.

We'll also look at how to estimate rock properties using different approaches, including deriving bulk properties from mineralogy. This approach introduces mixing models, and we'll discuss when different mixing relationships are appropriate.

What to submit (Week 1 practical; total 20 points)

A. **Gradient worksheet (10 pts).** Submit a scan/photo/PDF of your completed sheet (arrows, notes, short answers). This will be done as a group, but submitted individually. You may all work off the same worksheet.

B. **Programming (10 pts).** Submit your notebook or `.py` with *your own* 5-point Laplacian, plus: a regional map, a $\nabla^2 h$ map with symmetric limits, and a 3–5 sentence interpretation (why $|\nabla^2 h|$ highlights sharp features and why increased distance smooths).

Rubrics (concise)

Gradient worksheet (10): correctness of arrows (4); thoughfullness of written answers (4); accuracy of written answers (2)
Programming (10): code correctness (2); interpretation (8: thorough and thoughtfull answers to questions).

Tips

- Work in mixed-background groups; geologist + physicist + ...
- Share your expertise freely and use each other's knowledge to fill gaps in your own.
- Work colletively and constructively towards the common aims. Value each other's opinions, feel free to respectively discuss differences of ideas.
- Keep your regional window modest for the computer exercises so computations render quickly.

Workshop Notes

Workshop Notes

Activity A
Tectonic Plate vs. Freight Train

1 Purpose

The purpose of this activity is both a warm-up and an introduction to back-of-the-envelope (BOTE) problem solving. So called a BOTE calculation because it is one that can be done relatively simply on a small piece of scrap paper. Such calculations often result in order-of-magnitude estimates, are a way to place rough bounds on a value, and can be used test simple hypotheses before more complex computations are required. It is a useful skill because it can be done in the field to do simple interpretations of data or during a chat with colleagues over drinks/dinner.

2 How to do BOTE calculations

Work backwards. Start with the last equation you will need to solve the problem. Write down what you know and identify what you don't. Often you will need to do multiple calculations on the way to solving the problem.

A couple things to keep in mind:

round quantities – we are generally looking for order-of-magnitude estimates, not precise answers;
estimate unknowns – few values are truely unknown, bounds even rough can often be set;
use geometric means – you will find that when using bounds values tend towards the smaller bound, a geometic mean $g = \sqrt{(}\text{lower} \times \text{upper})$ of the bounds is often closer to the truth than an arithmetic mean;
use scientific notation – quicker for many unit conversions, multiplication and division;
dimensional analyis – always check your units, it can reduce the number of mistakes and can be a way to construct equations when you can't remember the exact formula

2.1 Rounding

We can round numbers to make calculations quicker and easier. For example, how many seconds are there in a year? The exact answer requires the number of seconds in a minute, the number of minutes in an hour, the number of hours in a day, and the number of days in a year (don't forget leap years).

That's too complicated. The number is approximately 3.15×10^7 s. There are two values I keep in my head, $\pi \times 10^7$ s and 3×10^7. The former is useful because π shows up in a lot of equations and this is often an easy way to cancel it, and the later is rounded.

2.2 Bounds

Even when we have little information about the value for a property or a quantity, we can place some rough bounds if we can't compute it from an equation where we have better constraints on all the terms. For example, we know the velocity of a tectonic plate is between 0 and some value. We can easily outwalk it so we know it is less than m s^{-1}. You might have heard that it is about the rate that fingernails grow. Some will be faster and some slower, so I might choose a value between say 10 and 100 mm a^{-1}. Many natural values are log-normally distributed, which means they are skewed towards lower values but have long tails on the high end. In such cases it makes more sense to use a geometric mean to calculate an average than an arithmetic mean. The geometric mean of two values is simply the square root of their product, in our plate velocity example,

$$v_g = \sqrt{(10\ \text{mm a}^{-1})(100\ \text{mm a}^{-1})} \qquad = \sqrt{(1000\ \text{mm a}^{-1})} \approx 30\ \text{mm a}^{-1},$$

which in this case turns out to be about right.

2.3 Scientific notation

Scientific notation converts every number into a decimal multiplied by a power of 10, i.e., $m \times 10^n$, where m is a decimal value and n is an integer. The number m is called the mantissa or significand and n is called the order or exponent. When multiplying two numbers in scientific notation, we multiply the mantissa and add the orders,

$$(m_1 \times 10^{n_1})(m_2 \times 10^{n_2}) = (m_1 \cdot m_2) \times 10^{n_1+n_2}.$$

Division likewise is easier,

$$\frac{m_1 \times 10^{n_1}}{m_2 \times 10^{n_2}} = \frac{m_1}{m_2} \times 10^{n_1-n_2}.$$

Because we often only care about order-of-magnitude rather than precise estimates, we can limit our m_1 and m_2 to one to two digits.

Many unit conversion become a sinch in scientific notation. Using our seconds in a year above, if we needed the number of seconds in a million years (as geoscientists we often work in deep time) we multiply by 10^6. Hence, $(3 \times 10^7\ \text{s a}^{-1})(10^6\ \text{a Ma}^{-1}) = 3 \times 10^{13}\ \text{s Ma}^{-1}$.

2.4 Dimensional analysis as a test

Dimensional analysis is the process of checking units during mathematical operations to ensure that the correct units are determined in the end. For example, if you calculate the force of gravity from $F = mg$, how do you know that you end up with the appropriate units of newtons (N)? We can write this as an equation:

$$\begin{aligned} F &= mg \\ [\text{N}] &\stackrel{?}{=} [\text{kg}][\text{m s}^{-2}] \\ [\text{N}] &= [\text{kg m s}^{-2}] \end{aligned}$$

Of course, this can only be verified as correct if you know that a newton is defined as a kg m s^{-2}, so it is useful to know how units are defined in terms of base units like m, s, kg, C, etc.

Let's look at another example, the calculation of pressure. Such a calculation is commonplace for geoscientist. Often we report pressures in mega-pascals, MPa. The equation for pressure is simple,

$$P = \rho g h,$$

where ρ is the average density of rock above the point of interest, g is the gravitational acceleration, and h, is the thickness of rock above. Density is typically reported in kg m^{-3}, gravity in m s^{-2}, and height in km. Note neither pressure or height are reported in standard SI units. So let's see if this works dimensionally,

$$\begin{aligned} [\text{MPa}] &\stackrel{?}{=} [\text{kg m}^{-3}][\text{m s}^{-2}][\text{km}] \\ [\text{MPa}][\text{Pa MPa}^{-}1 \cdot 10^{-6}] &\stackrel{?}{=} [\text{kg m}^{-2}\ \text{s}^{-2}][\text{km}][\text{m km}^{-1} \cdot 10^{-3}] \\ 10^{-6}[\text{Pa}] &\stackrel{?}{=} 10^{-3}[\text{kg m}^{-1}\ \text{s}^{-2}], \end{aligned}$$

since a Pa is a N m^{-2} or a kg s^{-2} m^{-1}, we can see

$$10^{-6}[\text{Pa}] \neq 10^{-3}[\text{Pa}].$$

So, we are missing a conversion factor of 10^{-3} to give our units in MPa.

Question 1. Tectonic Plate versus Freight Train
For this problem, I want you to answer two questions:

(a) Which has larger momentum? Momentum is given by $p = mv$, where m is mass and v is velocity.

(b) Which has larger kinetic energy? Kinetic energy is given by $K = \frac{1}{2}mv^2$.

Remember to work backwards. Start with mass, how do you estimate it for a tectonic plate, a freight train? Google won't be much help because it cannot tell you the answer (and often gives one orders of magnitude off when students ask). You'll need to do some out of the box thinking, recall some things you might know and some things you might not, for instance, what comprises a plate, not just horizontally, but vertically?

You can use Google for some basic geology questions, but don't ask it for numbers/quantities.

Use this page to complete your BOTE estimates.

Activity B
Sheet Model of Potential Fields

3 Purpose

This short activity uses a physical sheet model to develop intuition for scalar fields, gradients, and curvature. By fixing boundary values and introducing localized perturbations, you will explore how fields respond to boundary conditions and internal sources, and how these effects relate to geophysical potential fields.

4 Setup

Work as a group around a stretched sheet. Some students will hold the edges, while others gently push or pull on the sheet from below or above.

5 Tasks and Prompts

Question 1. Boundary conditions (Dirichlet)
Hold the edges of the sheet at fixed but varying heights and pull it taut.

(a) How do the boundary values control the shape of the interior of the sheet?

(b) Are there any local maxima or minima within the interior?

Question 2. Sources and sinks (Poisson region)
Introduce localized perturbations by pushing up (source) or pulling down (sink) at points within the sheet.

(a) How does the interior of the sheet respond to these perturbations?

(b) Do these changes affect the boundary? Why or why not?

Question 3. Gradients and curvature
Consider the geometry of the sheet as a surface.

(a) Where is $|\nabla\Phi|$ largest? How can you tell from the shape of the sheet?

(b) Where is $\nabla^2\Phi \neq 0$ versus ≈ 0?

(c) Which features correspond to regions of strong curvature?

Question 4. Effect of observation distance
Imagine observing the sheet from farther above.

(a) Why does the surface appear smoother as the observation distance increases?

(b) Which features disappear first, and why?

6 Key idea

Gradients describe how quickly and in what direction a field changes, while curvature describes how those changes themselves vary in space. In geophysical potential fields, boundary conditions and internal sources control the field, and increasing observation distance smooths short-wavelength features.

Activity C
Gradients Activity

"I will never listen to ocean waves or view a beautiful sunset in quite the same way again. That is perhaps the greatest gift one can gain by delving into calculus: It is a whole new way of looking at the world, accessible only through the realm of mathematics."

— Jennifer Ouellette, *The Calculus Diaries: How Math Can Help You Lose Weight, Win in Vegas, and Survive a Zombie Apocalypse*

7 Introduction

As we study geophysical theory it will become clear that in order to unlock the secrets of the physical world, we must understand the changing nature of fields. If the fields were constant the world would be very boring indeed, for it would not contain any variations in physical properties that we could remotely sense. Variations in physical fields help us better understand the state (temperature, pressure, and composition), structure, and/or the evolution of our wonderous planet. One of the simplest ways to explore a field is to examine the rate of change of the field, or gradient, which is really a fancy way of saying slope.

In this workshop you will explore conceptually the physical meaning of a gradient using intuitive examples from elevation. We will be able to see how the gradient depends upon the direction we consider and how a derivative can be calculated simply for a complex geometry, and how the gradient of a scalar field such as elevation results in a vector field requiring both magnitude and direction in order to describe the gradient.

We will start the exercise by building a conceptual understanding of the gradient before creating a more complete mathematical description.

8 Mount Shasta Example

Figure 1: Mount Shasta, California (image from Google Earth).

Elevation is a good place to start to think about gradients because we all intuitively know what a slope means in this context. As far as topography goes, Mount Shasta is pretty simple. It is a stratovolcano at the southernmost end of the associated with the Cascade Volcanic Arc, and looks like a textbook example of a volcano in that it is fairly symmetric radially and sits in the middle of a vast flat plain.

Question 1. Shasta gradient

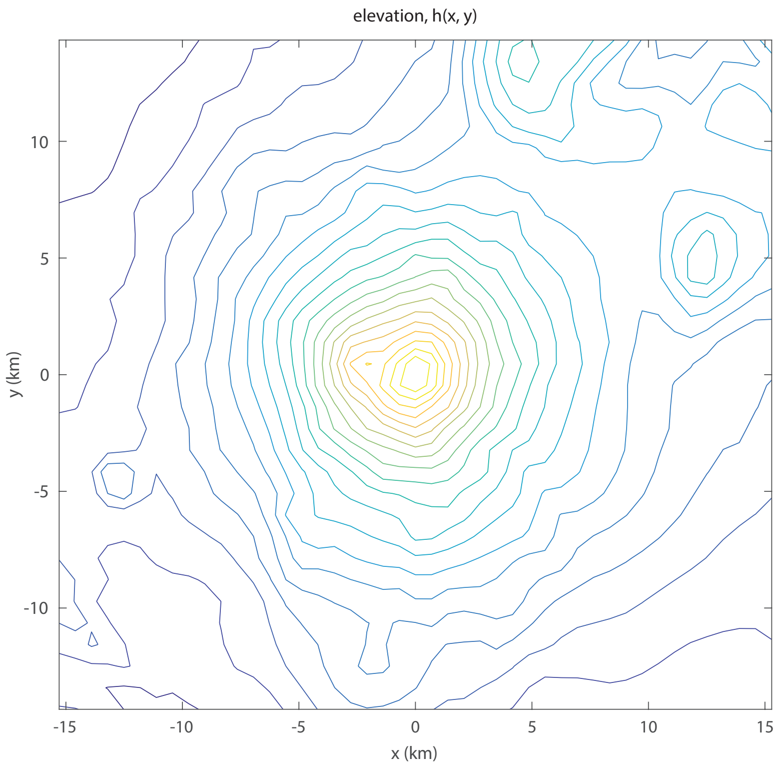

Figure 2: Topographic map of Mount Shasta.

(a) Over the topographic contours on Figure 2, draw a series of arrows pointing up slope representing the opposite direction water might flow. Make the arrows longer when the slope is steeper and make the arrows shorter when the slope is gentler.

(b) How did you decide to draw the direction of the arrows?

(c) How did you decide to draw the length of the arrows?

(d) Why does it require two quantities, length and direction, to describe the gradient?

9 The mathematics of gradients: the slopes of Mt. Shasta

Consider hiking a profile line over the peak of Mt. Shasta (Figure 3, top panel). You will start by walking up the slope of Mt. Shasta. The slope starts gently going up in elevation (a positive slope in Figure 3, bottom panel), which increases as you climb in elevation. The slope increases and reaches a maximum near the peak, then the slope starts to decrease as elevation still increases. Once you reach the snow-capped peak there is no change in slope (i.e., slope = 0). Then you decide to pull out your sled and quickly decend the other side (negative slope).

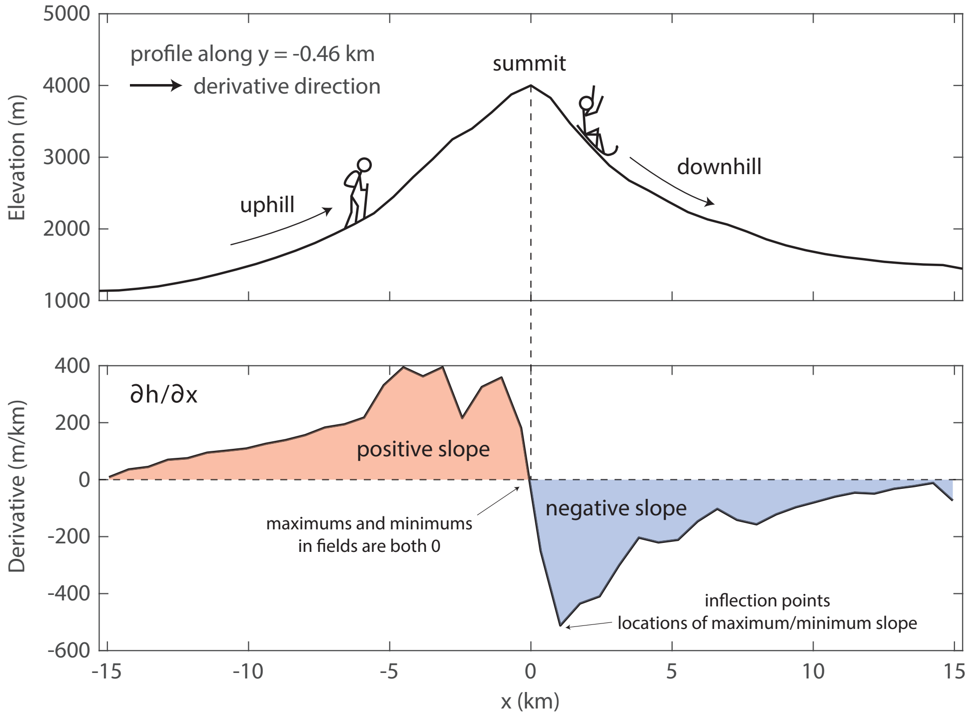

Figure 3: Elevation profile (top) and gradient along the elevation profile (bottom).

What we want to do is be able to describe the changes in slope that you as a hiker feels mathematically. Say every step you take is a similar distance laterally (the run) and the height you go up (or down, negative up) in elevation (the rise) allows us to express the slope as

$$\text{slope} = \frac{\text{rise}}{\text{run}}. \tag{1}$$

But the direction that you take matters. Since we don't necessarily know the direction of the maximum slope we'll force the hiker to walk in just one direction, say west to east ($+x$) to start. The size of the steps they take are Δx. In this case, we can express the slope as

$$\frac{\partial h}{\partial x} \approx s_x = \frac{h(x+\Delta x) - h(x)}{(x+\Delta x) - x} = \frac{h(x+\Delta x) - h(x)}{\Delta x}. \tag{2}$$

At this point the way we solve this mathematically as opposed to a computer are different. A computer will solve the problem much like above, taking the difference between the heights at $h(x+\Delta x)$ and $h(x)$ and dividing by the distance of each step (Δx).

For our purposes generally during the course, we will take a more mathematical approach, taking the case where we want to know the slope at a continous series of point along the elevation profile. In this case the distances between our steps approach zero and the slope we approximate the approaches a limit of the true slope at each point. This is the definition of the derivative, and you can find an extended discussion as a tutorial online and in the math appendicies in the course notes.

If instead, you were to hike from south to north ($+y$-direction) we we can write the slope as

$$\frac{\partial h}{\partial y} \approx s_y = \frac{h(y+\Delta y)-h(y)}{(y+\Delta y)-y} = \frac{h(y+\Delta y)-h(y)}{\Delta y}. \tag{3}$$

If you were to walk in a series of profiles in x or y, you could map out the slope with respect to that series of traverses. You will notice that the slope you get is very different depending upon whether you walked in the x-direction or the y-direction [1]. We can write the function of the slope, s at each location (x, y) as

$$\mathbf{s}(x, y) = s_x\hat{\mathbf{x}} + s_y\hat{\mathbf{y}}, \tag{4}$$

where the $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ are a unit vectors in the x- and y-directions, respectively. Note that the slope is a vector quantity, which means it has both a magnitude and a direction. In calculus, we describe the gradient ($\boldsymbol{\nabla}$) of the elevation function, $h(x, y)$, by

$$\boldsymbol{\nabla} h = \frac{\partial h}{\partial x}\hat{\mathbf{x}} + \frac{\partial h}{\partial y}\hat{\mathbf{y}}. \tag{5}$$

Because the components are in different directions, we cannot simply add them to find the maximum slope.

From the result above, we learn that the slope is different depending upon the direction you walk (N-S) versus (E-W) which confounds the direction of the maximum slope. You may also note that if we had walked in the opposite (negative) directions, we would have changed the sign of our slopes. Do not fear, there is a simple mathematical solution to determine both the maximum slope and its direction.

If we assume that from each point that we walked in either the +x or +y direction we can describe the local surface as a plane, the directions we walked were purposfully perpendicular to each other. It turns out that these two directions can be used to define that plane and both the angle and magnitude of the maximum slope (magnitude) along it (Figure **??**). To determine the maximum slope along each individual little plane, we use the Pythagorean theorem, which you may remember from high school,

$$\|s\| = (s_x^2 + s_y^2)^{1/2}. \tag{6}$$

Note that because we square each slope component before adding them, we have ensured that our resultant magniude of the slope is positive. To determine the direction of the maximum slope, we can use trigonometry to determine the angle with respect to the x-axis as

$$\tan\theta = \frac{s_y}{s_x}. \tag{7}$$

Note that to find the appropriate angle we will have to keep track of the signs of s_x and s_y to ensure that θ lies in the appropriate quadrant.

10 Marble Canyon Example

Let's look at another topographic example at Marble Canyon in Arizona. Marble Canyon sits before the entrance of the Grand Canyon. The area surrounding the canyon has relatively flat topography except for some steep-sided cliffs, one east of the canyon that runs approximately north to south and a second, north of the canyon which varies in azimuth. The canyon itself runs northeast to southwest though the center of the image, carved by the Colorado River.

In this exercise, we will take the previous example one step further by determining the slope along each individual axis and then combine them as we might in the mathematical sense and then combine the results to obtain determine a magnitude and direction. Again, we'll do this graphically rather than mathematically to

[1] Since Mt. Shasta is fairly radially symmetric, you will notice that the results are pretty similar if you rotate the one of the plots 90° with respect to the other. This is a relatively special case as we can't generally do this, however, many of the problems we will solve in this course either change in one direction only or are symmetric in some way to make the problems simpler.

Figure 4: Marble Canyon, Arizona (image from Google Earth).

gain further insight into how the mathematics works. Note there is a substantial region where the topography is relatively flat (i.e., the slope is negligible). You don't need to put any arrows in these positions so focus on regions where elevation changes.

Question 2. Marble Canyon

(a) In Figure 5 are two of the same topographic map of Marble Canyon. On the top map, make a series of arrows parallel to the x-axis with length represpresenting the slope in the x-direction. Again, arrow should point up slope. Now on the same topographic map, make a series of arrows parallel to the y-axis with length representing the slope in the y-direction. Once you have both sets of arrows, create a new set of arrows on the bottom topograpic map that represent a combination of the two. The resultant vector is simply a graphical application of the Pythagorean equation. What you will have done is the same thing that the math is doing. So when we are working with abstract fields that cannot be seen or felt by us, remember, the concept is shown here is the same.

(b) How do the individual derivative components behave with respect to the ridge running north east of the canyon?

(c) For the northern ridge, why does the y-component have only positive slopes whereas the x-component is positive in some places and negative in others?

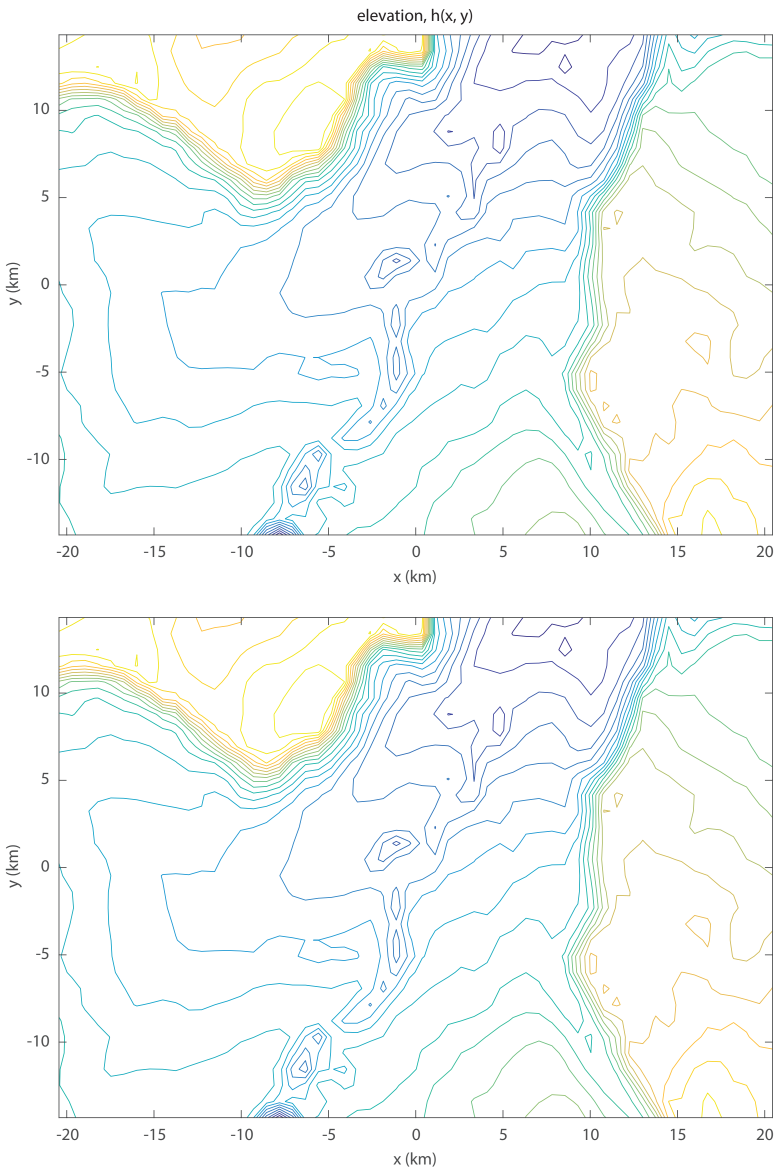

Figure 5: Topographic map of Marble Canyon.

(d) Can you find examples of either of the above situations occuring in the canyon itself?